



Robotics Discovery Program

Ranger Bots Modular Workshop Series

[New England Sci-Tech • WPI IQP Project • June 2026]





NEW ENGLAND
SCI-TECH



WPI

Hi, I'm Jefferson!

About Me

I like video games, drawing, sports, wandering around, and listening to music way too much. I guess robots are cool too.



ROBOTS

Studying WPI Robotics & ECE



DESIGN

Graphic design + Drawing



TEAM

Wrestling + SOMA

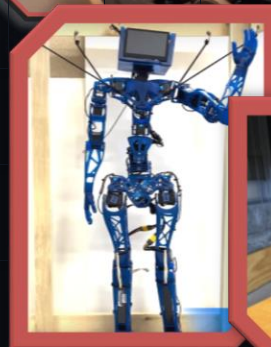


MINDSET

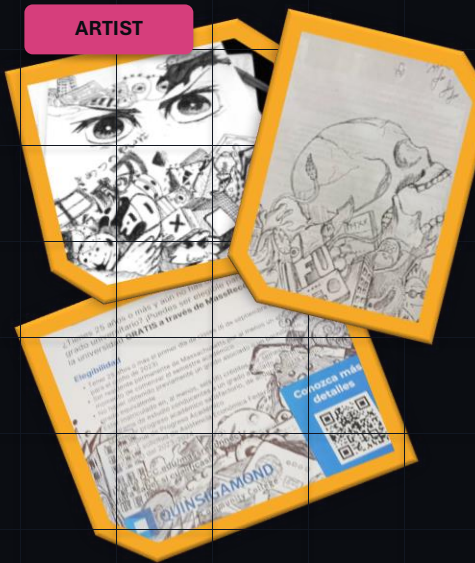
Give it your best

It Fun To Be a Jack Of Trades

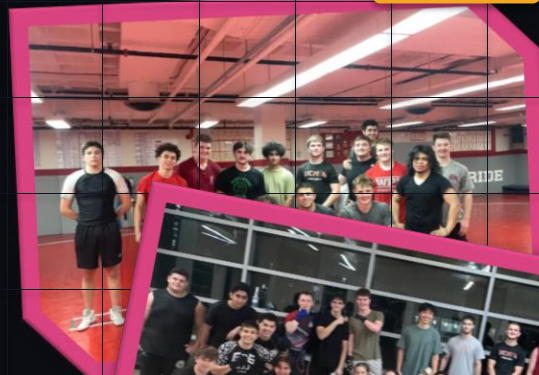
ROBOT



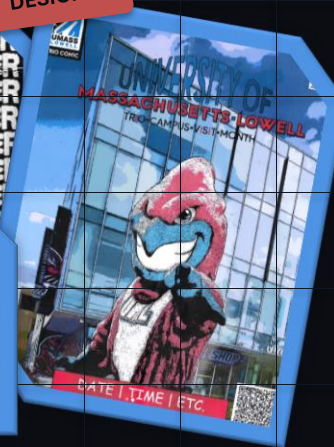
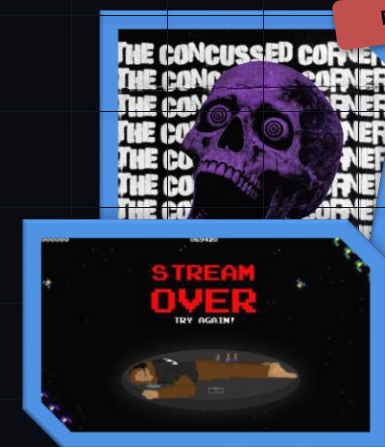
ARTIST



TEAM



DESIGN

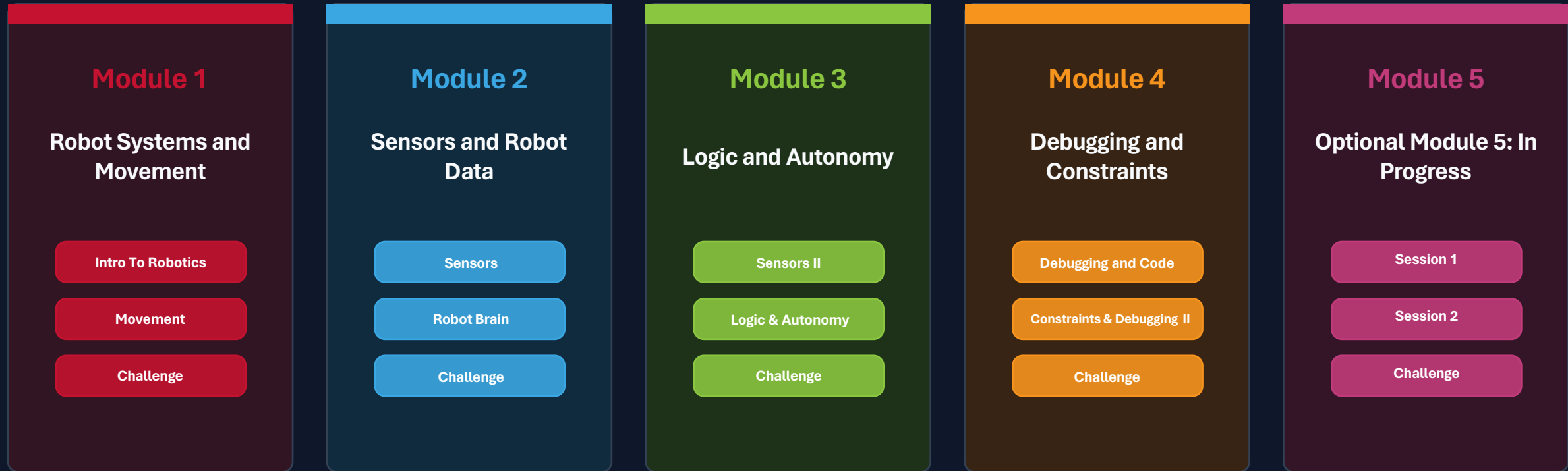


Your Turn

What is your name?
and
What is your favorite ____?

Modular Workshop Map

Each module has Session 1, Session 2, and a Challenge. Module 5 is optional if the extra week happens.



FIND YOUR DUO

Robot teams today are pairs.

Not solo. Not a huge squad. Just you, a teammate, and one robot mission.

STOP

- 1 Find one partner
- 2 Say your name
- 3 Pick a team name

Famous duo energy



Your duo job: Collaborate with each other



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Module 1: Session 1

Introduction To Robotics

BIG IDEA

Robots are systems: body, brain, sensors, and code working together.

ASK

What makes something a robot?

SHOW

mBot Ranger parts: body, wheels/treads, motors, sensors, battery, and controller.

TRY IT

1. Draw a robot that solves a problem.
2. Then redesign it when new limits are added.

MATERIALS

Paper, pencil, markers, mBot Ranger demo

CHECKPOINT

Can your team explain what your robot moves, senses, and decides?

Exit Question: What changed when your robot had limits?

ROBOTS COME IN MANY FORMS

They all have something in common



1. Humanoid

Body + sensors



2. Self-driving car

Sensors + decisions



3. Legged robot

Balance + motion



4. Home robot

Autonomy at home



5. Drone

Flight + control



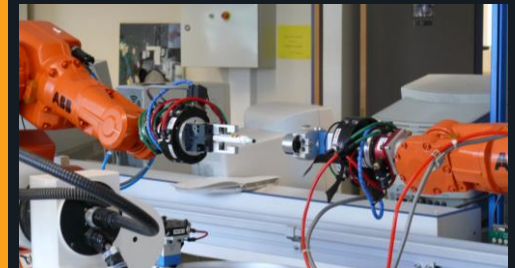
6. Student robot

Build + test



7. Surgical robot

Precision tools



8. Factory arm

Repeatable action

Ranger Bots starts small, but the ideas scale up.

Movement • sensors • logic • debugging • teamwork

A ROBOT USUALLY HAS

Four important parts

Hey look at your Robot !!!

1

Body

The robot has a body and works in real space.

2

Sensor

It gathers information from the world.

3

Actuators

It moves or affects something around it.

4

Brain

It is trying to complete a mission.

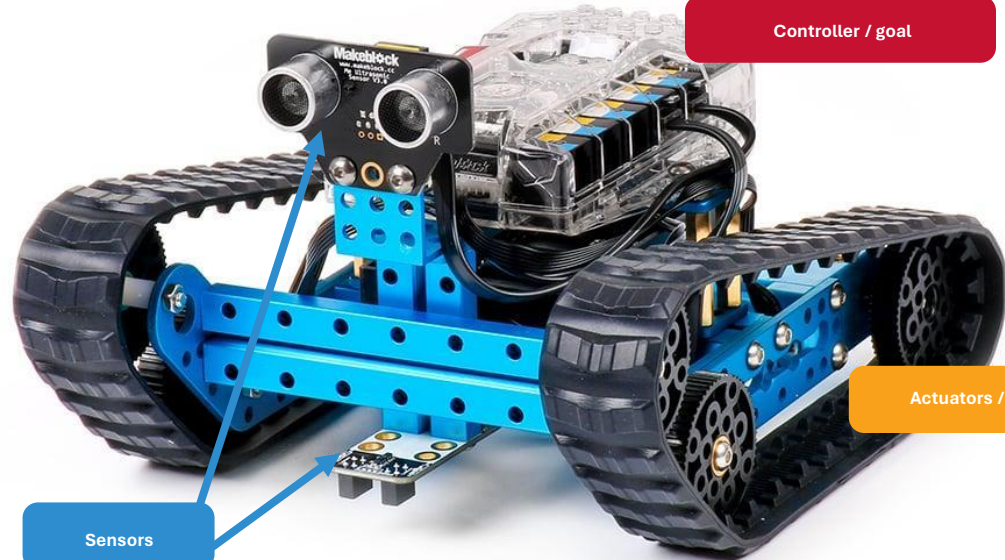
Quick check: point to each part on the robot.

Body / physical robot

Controller / goal

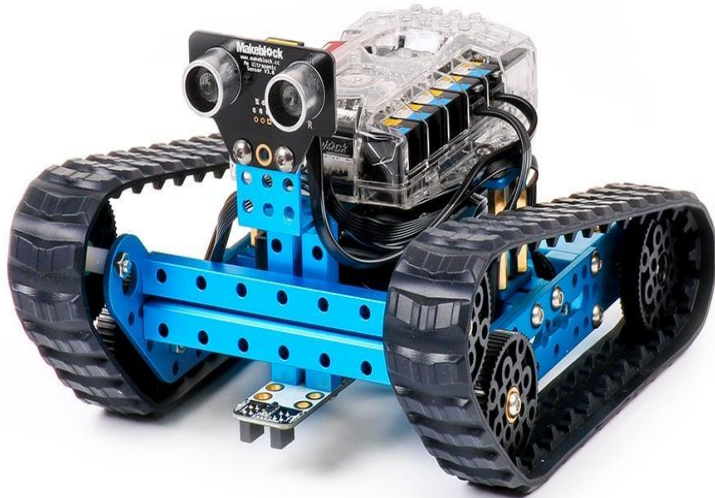
Actuators / Motors

Sensors



ROBOT PARTS WORK TOGETHER

Sense → Think → Act → Improve



How does your robot work?

Brain • Sensors • Actuators • Power • Frame

Sensors

Gather clues: distance, light, line, touch, motion.

Brain

Reads the data and chooses the next command.

Actuators

Motors, wheels, arms, or parts that create action.

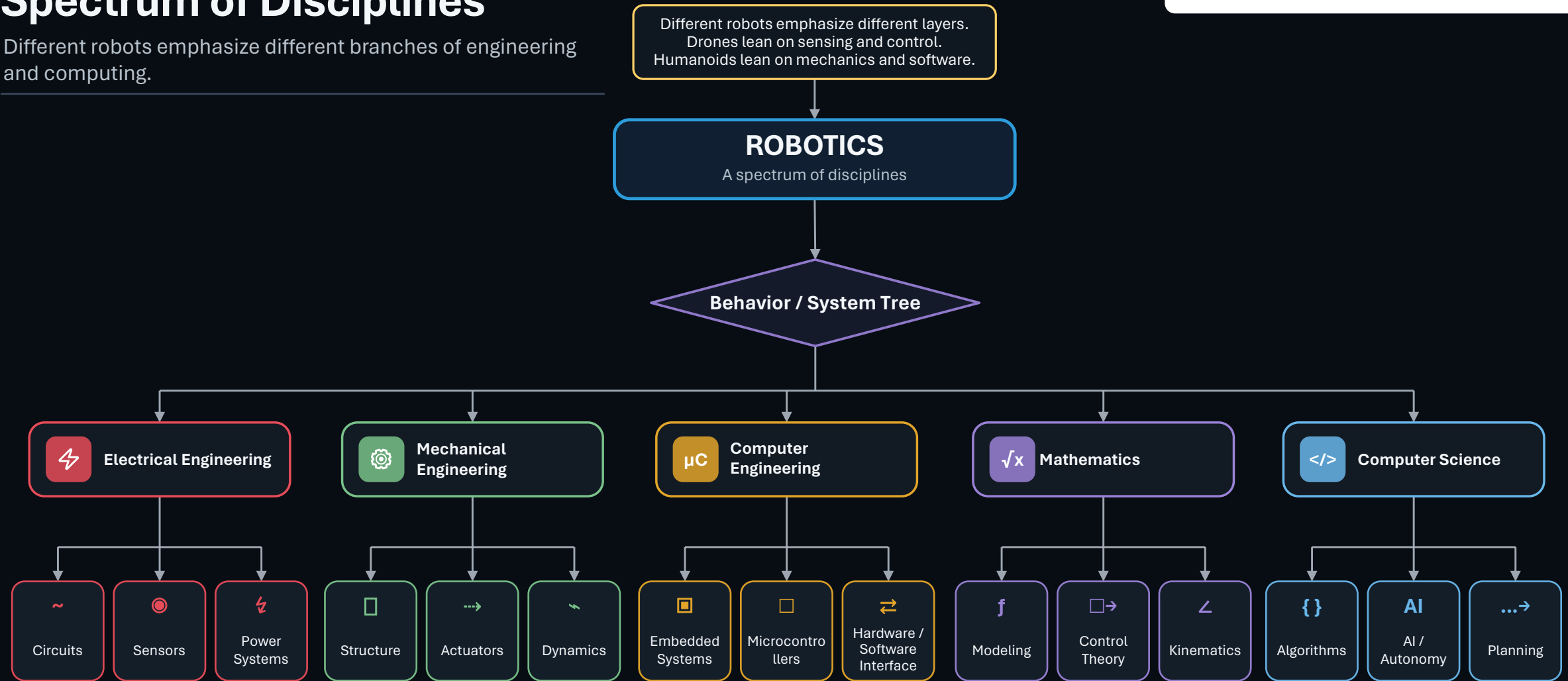
Goal

Accomplish a set of Task efficiently

Fun Fact : The word robot comes from the Czech word robota, meaning forced labor or compulsory work

Spectrum of Disciplines

Different robots emphasize different branches of engineering and computing.



Draw A Bot

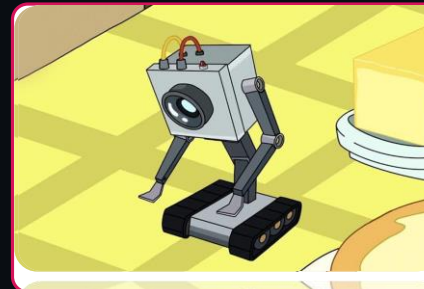
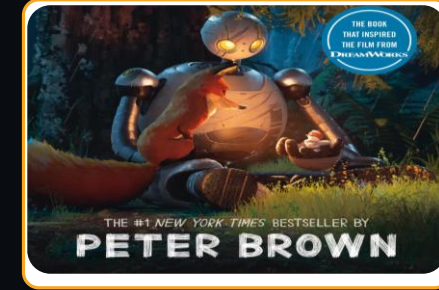
What Are The RULES

Adding the A back in STEM

Creativity isn't just making something new. It's solving something differently.

- 1 Given A Setting, Constraints, And A Goal
- 2 Draw a robot to accomplish the goal
- 3 Present your robot

Examples



Your duo job: Collaborate with each other

DRAW-A-BOT WORKSHOP – Round I



CONDITIONS A

NONE

CONDITION B

NONE AGAIN

CONDITION C

NONE TIMES 3

DRAW-A-BOT WORKSHOP – Round II



CONDITIONS A

Robot may not fly

CONDITION B

The robot must avoid bumping into people

CONDITION C

The robot can only carry one item at a time.

DRAW-A-BOT WORKSHOP – Round III



CONDITIONS A

Work in the dark

CONDITION B

Deal with blocked path

CONDITION C

Must know when it has a fragile item